



DATA SCIENCE LAB-1 [MTH208]

Project Proposal

Advanced Stock Market Dashboard using R

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1. Problem Statement and Objectives

Current market dashboards such as TradingView and others provide stock visualizations, charts, and indicators, but many lack a combination of *statistical trend prediction, integrated fundamental ratios, and custom technical strategy visualization.* Our project aims to develop an **R Shiny-based advanced stock market dashboard** with the following key features, differentiating it from existing platforms:

Dashboard Features:

- Display of stock prices and interactive charts.
- Technical indicators (MA, RSI, Bollinger Bands) for price trends.
- Integration of fundamental company ratios for deeper insights.
- Market trend predictions (Bullish/Bearish) for upcoming days using statistical indicators derived from historical data.

2. Planned Data Sources and Feasibility & Ethics Check

Data Sources:

- NSE India website, Yahoo Finance APIs for OHLCV and historical data.
- R packages: `quantmod`, `tidyquant`, `rvest`, `TTR`, `ggplot2`, `plotly`, `shiny`.
- Backtesting modules using `PerformanceAnalytics` and `quantstrat` for portfolio evaluation.

Feasibility: Backtesting integration ensures fast evaluation of trading strategies, interactive dashboard creation, and reproducibility. Packages support visualization, statistical analysis, and portfolio modeling.

Ethics Check: Only publicly available data will be used; no proprietary or confidential data accessed. Academic project purposes only.

3. Proposed Analysis Plan

1. **Web Scraping / Data Retrieval:** Fetch OHLCV stock data using APIs and R scraping libraries.
2. **Data Cleaning and Preprocessing:** Handle missing values, normalize formats, remove outliers.
3. **Exploratory Data Analysis (EDA):** Compute mean, std, max-min, range, returns, volume analysis, distribution plots using `ggplot2`.
4. **Technical Analysis and Dashboard:** Candlestick charts, overlay MA, RSI, Bollinger Bands, market trend prediction (Bullish/Bearish) using statistical indicators; interactive charts in R Shiny.
5. **Fundamental Analysis Integration:** Add company ratios (P/E, ROE, Debt/Equity, etc.) for

comprehensive insight.

6. **Portfolio Optimization:** Implement Markowitz mean-variance allocation, CAPM expected returns; compute optimal weights and perform backtesting strategies.
7. **Options Trading Analysis (Optional / Time Permits):** Black-Scholes-Merton model for European option pricing; compute Greeks for risk assessment.

4. Timeline and Work Distribution

Date / Week	Tasks
7 Oct	Project proposal submission; finalize objectives and data sources.
8-14 Oct	Web scraping and API data collection; preprocessing and cleaning.
15-21 Oct	Exploratory Data Analysis (EDA), compute statistics, visualize distributions.
22-28 Oct	Technical indicators (MA, RSI, Bollinger Bands) and R Shiny dashboard development; integrate statistical trend predictions.
29 Oct - 1 Nov	Complete dashboard, portfolio optimization, backtesting, integrate fundamental ratios and options analysis.
2 Nov	Final submission and presentation; demo dashboard, trend prediction, portfolio model, options strategies.

5. Tentative Team Roles

- **Suyash Jindal & Rachit Jindal:**
 - Web scraping, API data retrieval.
 - Technical indicators and backtesting implementation for performance evaluation.
 - Dashboard interactivity, plotting, and visualization.
 - Portfolio optimization computations and strategy evaluation.
 - Options pricing using Black-Scholes-Merton and computing Greeks.
- **Vijay Jadhav & Venkatesh Tejavath:**
 - Exploratory Data Analysis (EDA) and statistical summaries.
 - Dashboard layout, fundamental ratios integration, testing, and usability.
 - Documentation, report preparation, and presentation.